Max Hartman

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Education

University of Illinois **Champaign, IL**

M.S. in Electrical & Computer Engineering *2025 – 2027*

University of Illinois **Champaign, IL**

B.S. in Computer Engineering, Minor in Mathematics *2022 – 2025*

Advisor: Lav Varshney

Thesis: SparseJEPA: Sparse Representation Learning of Joint Embedding Predictive Architectures

GPA: 3.91/4.00 (Graduated with Highest Honors)

Research Areas

Interpretability & AI Safety, Vision Language Models, Representation Learning, Machine Learning, Computer Vision, Federated Learning

Publications

Large Language Model Interpretability Methods are not Actionable for Multimodal Models. Vidhata Jayaraman*, **Max Hartman***, Moulik Choraria, Akhil Bhimaraju, Lav R Varshney, 2025. In Review at ICML Workshop.

SparseJEPA: Sparse Representation Learning of Joint Embedding Predictive Architectures. **Max Hartman,** Lav R Varshney, 2025. Preprint.

SwitchCIT: Switching for Continual Instruction Tuning of Large Language Models. Xinbo Wu, **Max Hartman**, Vidhata Jayaraman, and Lav R. Varshney, 2024. Preprint.

* Denotes Equal Contributor

Industry Experience

Amazon (AWS) Seattle, WA

Software Development Engineer Intern Summer 2025

Electrokare Champaign, IL

Founding Engineer 2023 – 2024

Honeywell International Atlanta, GA

Artificial Intelligence/Machine Learning Engineer Intern — Summer 2024

Genuine Parts Company — Atlanta, GA

Software Development Engineer Intern — Summer 2023

Teaching

Intro to Computer Engineering (UIUC ECE 120)

Undergraduate Course Assistant — *Spring 2023 - Spring 2025*

Mentorship and Outreach

SIAM/MathWorks Math Modeling Challenge

Problem Development Committee, Judge — *2026 - Present*

Support SIAM's annual math modeling competition, which gives high school students an opportunity to apply mathematics, statistics, and data science to a real-world problem. Serve on committee in charge of writing practice and contest problems. Judging student submissions.

Honors and Awards

University of Illinois (Undergraduate)

Engineering Visionary Scholarship, 2024, $2,000

Illinois Engineering Outstanding Scholarship, 2023 & 2024 — $2,000 (Each)

Schlader Memorial Scholarship in Engineering, 2023 — $3,000

Society for Industrial and Applied Mathematics

MathWorks Math Modeling Challenge Magna Cum Laude, 2022 — $15,000